13.4 Generating Equivalent Algebraic Expressions

Lesson Introduction

Benchmark	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.	Learning Target	• I can apply properties of operations to generate equivalent expressions.

Benchmark Coherence	Building On	Working Toward
	MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.	MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Essential Questions	• How can you use properties to generate that two algebraic expressions are equivalent?
Lesson Narrative	This lesson begins by using the area model, which students are familiar with, to model the distributive property. The lesson primarily focuses on applying the properties of operations (associative, commutative, and distributive) reviewed in the previous lesson to generate equivalent expressions.

Component Summary				
	13.4.1 Warm-Up (~5 min.)	Analyze a puzzle for a pattern *(SE p. 280)*	13.4.3 Try It (5 min.)	Apply properties of operation to generate equivalent algebraic expressions *(SE p. 282)*
	13.4.2 Guided Instruction (20 min.)	Review, extension, and application of properties to generate equivalent expressions *(SE p. 281)*	13.4.4 Your Turn! (10 min.)	Additional practice applying of properties of operation to generate equivalent algebraic expressions through area model and matching exercises *(SE p. 283)*
	13.4.5 Wrap-Up (~5 min.)	Demonstrate understanding of applying properties of operation to generate equivalent algebraic expressions through error analysis *(SE p. 284)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Distributive property • Associative property • Commutative property	• (Optional) Colored pencils	• (Optional) Determine student groupings before class • (Optional) Prepare sentence stems, if needed

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not distribute the term outside of the parentheses to all terms in the parentheses, or students may distribute the term outside the parenthesis to all terms in the expression regardless of where they are.
Students may incorrectly apply properties to contexts with subtraction and division.</td><td>Properties are not new information, but not all students will be entering this lesson on the same level. Consider what information can be used prior to this lesson to better inform groups and differentiate throughout.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Engage students in a Think-Pair-Share routine during 13.4.4 Your Turn!. Ask students to first complete question 2 on their own, then share their responses with a partner and subsequently make any changes to their expressions before sharing out as a class. This routine encourages discourse and solidifies understanding.</td><td>MA.K12.MTR.5.1: Patterns and Structures
In 13.4.4 Your Turn!, students use the structure of the expressions and properties to determine if different expressions are equivalent. Students also have to logical order their ideas answer the questions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Reflection and error analysis are positive contributors to self-regulation, math confidence, and conceptual understanding. However, not all students reflect and analyze the same way. Consider differentiating the ways in which students can engage in demonstrating their progress and mastery in this lesson.
13.4.2 Guided Instruction: Comparing Makayla’s and Soledad’s approaches with a partner
13.4.4 Your Turn!: Articulating their reflection and progress verbally with a partner or telling a family member at home and recording their conversation to submit as homework
13.4.5 Wrap-Up: Recording themselves using an app like FlipGrid, explaining their answer to someone at home or a friend in a different math class</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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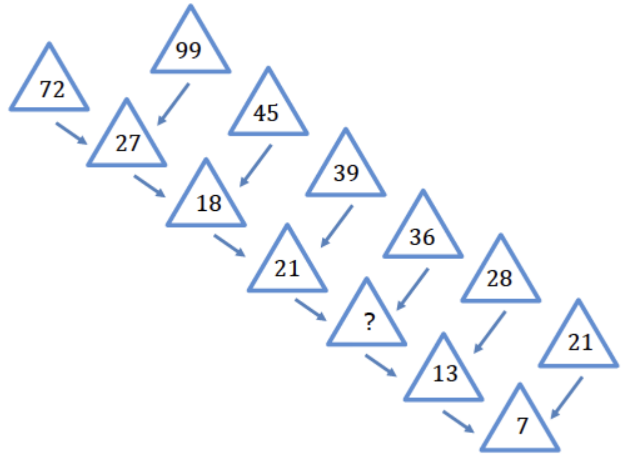
13.4.1 Warm-Up: Missing Number

Component Overview: *Students analyze a puzzle for a pattern that can be used to determine missing numbers.*

13.4.1 Warm-Up: Missing Number



1. What is the first number you tried for the missing number?
   Answers will vary.

2. Did your first number work for the rest of the pattern?
   Answers will vary.

3. What strategy did you try if your first number did not work?

   Add the digits. $7+2+9+9=27$
   $2+7+4+5=18$
   $1+8+3+9=21$
   $2+1+3+6=12$
   $1+2+2+8=13$
   $1+3+2+1=7$

Students may see the following pattern:
$99-72 = 27;\ 45-27 = 18$
This pattern works until the final step.

If this error is observed, prompt students to change the way they are thinking about operations and instead pay attention to the digits of each value, using an arrow to help make connections.

- *What do you notice about the relationship between the numbers? Does that work for every number?*
- *How else can you use the digits to determine the missing number? What strategy did you use?*
- *Is guess and check an efficient strategy?*

13.4.2 Guided Instruction: Generating Equivalent Algebraic Expressions

Component Overview: *Students use the properties of operations to write equivalent expressions. Students have several opportunities to practice applying the commutative, associative, and distributive properties. Students also check two samples of student work to see if they correctly applied the properties to write an equivalent expression by substituting a variable into each expression.*

13.4.2 Guided Instruction: Generating Equivalent Algebraic Expressions

1. Consider the following expressions.

   $8 \cdot x + 8 \cdot 9$ $\quad$ $8(x + 9)$ $\quad$ $8x + 9$

   a. Which of these expressions are equivalent?
   $8 \cdot x + 8 \cdot 9$ *and* $8(x + 9)$

   b. Which property of operation justifies that the expressions are equivalent?
   Distributive

2. Write two equivalent expressions. Name the property or properties you used to create each equivalent expression.
   $8 \cdot x + 9 \cdot 8$ *Commutative*
   $8(9 + x)$ *Commutative*
   $8 \cdot 9 + 8 \cdot x$ *Commutative*
   $x \cdot 8 + 8 \cdot 9$ *Commutative*

3. For each expression, use the associative and commutative properties to generate an equivalent expression.

Original Expression	Associative Property	Commutative Property
$(3x + 5) + 4z$	$3x + (5 + 4z)$	*Answers will vary* $4z + (3x + 5)$ $(5 + 3x) + 4z$
$8 \cdot (-1 \cdot 13d)$	$(8 \cdot -1) \cdot 13d$	*Answers will vary* $(-1 \cdot 13d) \cdot 8$ $8 \cdot (13d \cdot -1)$

4. Rewrite $-11b + 3(10k + 1)$ using the distributive property.
   $-11b + 30k + 3$

Focus attention on assisting students with writing equivalent algebraic expressions. The goal of this section is for students to understand there are many ways to rewrite expressions using the properties. Encourage students to think flexibly, particularly when writing expressions using each property.

- *How do* $(8 + 9) + 7 + 4 = 7 + (8 + 9) + 4$ *and* $(9 + 8) + 4 + 7$ *both demonstrate the commutative property?*
- *How do* $(8 + 9) + 7 + 4 = 8 + 9 + (7 + 4)$ *and* $8 + (9 + 7) + 4$ *both demonstrate the associative property?*
- *Which properties does* $(8 + 9) + 7 + 4 = (8 + 4) + (7 + 9)$ *show?*
- *Is it possible for expressions to demonstrate multiple properties?*
- *When you are rewriting expressions using the properties, is there only one way to do so?*

13.4.2 Guided instruction: Generating Equivalent Algebraic Expressions (continued)

5. Rewrite $-11b + 3(10k + 1)$ using the commutative property of multiplication.

   $-11b + (10k + 1) \cdot 3$ *or*
   $b \cdot -11 + 3(10k + 1)$ *or*
   $-11b + 3(k \cdot 10 + 1)$

6. Rewrite $-11b + 3(10k + 1)$ using the commutative property of addition.
   $-11b + 3(1 + 10k)$ *or*
   $3(10k + 1) + (-11b)$

7. Makayla and Soledad both rewrote $(-5h + 9) \cdot -14$ using the commutative property of multiplication. Verify that their expressions are equivalent by substituting a value in for h. Show your work in the table.

Makayla	Soledad
$-14(-5h + 9)$	$(h \cdot -5 + 9) \cdot -14$
$h = 1$	$h = 1$
$-14(-5(1) + 9)$ $-14(-5 + 9)$ $-14(4) = -56$	$(1 \cdot -5 + 9) \cdot -14$ $(-5 + 9) \cdot -14$ $4 \cdot -14$ $= -56$

Engage students in a discussion to provide an auditory entry point and check for understanding.

- *Why must the variable h be the same value in both Makayla's and Soledad's expressions?*

The benchmark MA.6.AR.1.4 states that students are to generate equivalent expressions. Note that the word "simplify" is not used in the instructions. Equivalence is different and has more useful mathematical applications than "simplifying."

13.4.3 Try It: Writing Equivalent Algebraic Expressions

Component Overview: *Students practice writing equivalent algebraic expressions.*

13.4.3 Try It: Writing Equivalent Algebraic Expressions

1. Select the expressions that are equivalent to $(2x+7)+(5y-3)$.

- ☐ $(5y-3)+(7+2x)$
- ☐ $(3-5y)+(7+2x)$
- ☐ $7+(5y+2x)-3$
- ☐ $(7+5y)+(2x-3)$
- ☐ $5y+(7+2x)-3$

2. Write two equivalent algebraic expressions for each expression below.

a. $(4+9)\cdot x$

$13x$

$x(4+9)$

b. $3t\cdot 4$

$12t$

$t\cdot 3\cdot 4$

c. $-7x+3-14$

$-7x-11$

$-7x+(3-14)$

d. $3(^{-}9p-4)+7$

$-27p-5$

$-27p-12+7$

Consider having students work in pairs or small groups to complete this component. Through discussions, students will formalize their understanding of using the properties of operations to write equivalent expressions.

Have students rewrite the algebraic expressions before and after applying the properties of operations using different colored pencils so that they can see the parts of the expressions that were changed.

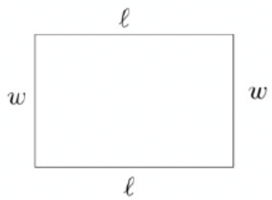
13.4.4 Your Turn!

Component Overview: *Students work independently to evaluate student work to determine if the expressions created accurately represent the area of a rectangle and then write to the student explaining why their expression is not equivalent to the others. Students then match equivalent expressions in a table, creating equivalent expressions for those who do not have matches.*

13.4.4 Your Turn!

1. The students in Ms. Justino's class are writing expressions for the perimeter of a rectangle of side length, ℓ, and width, w. The expressions for four students are in the table.

Student	Expression
De'Andre	$2(\ell + w)$
Summer	$\ell + w + \ell + w$
Abm	$2\ell + w$
Luzarie	$2\ell + 2w$



Ms. Justino told the group that one of the expressions is not equivalent. Write a short explanation to the student to show why their expression is not equivalent to the other expressions.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Abm – Your expression is not equivalent to

$2(l + w)$, $l + w + l + w$, *and* $2l + 2w$.

$l + w + l + w = 2l + 2w$

Your expression does not have the same coefficient for w.

Providing sentence stems will help students who may struggle with written responses. Additionally, allowing students to pair up and write joint responses may elicit more precise responses.

13.4.4 Your Turn! (continued)

2. Match equivalent expressions by drawing an arrow. If an expression does not have a match, write an equivalent expression to match it in one of the blank boxes provided.

Column A	Column B
$-8(x+3)$	$2+(-8x+3)$
$-8x \cdot 3$	*Answers will vary* $2(8x-3)$ $(-3 \cdot 8x) \cdot 2$
$(8x \cdot -3) \cdot 2$	$3 \cdot -8x$
$(3+-8x)+2$	$-24-8x$
$2+\ -8x$	*Answers will vary* $-8x+2$ $2-8x$

Engage students in a Think-Pair-Share routine. Ask students to first complete question 2 on their own, then share their response with a partner and subsequently make any changes to their expressions before sharing out as a class. This routine encourages discourse and solidifies understanding.

- *How can you find an equivalent expression?*
- *Did you and your partner create the same expressions? If not, are they both equivalent? How do you know?*
- *How can you use the properties of operations to find and create equivalent expressions?*

13.4.5 Wrap-Up: Error Analysis

Component Overview: *Students analyze a sample of student work to identify any errors in writing equivalent expressions. Students explain the error and correct the mistakes.*

13.4.5 Wrap-Up: Error Analysis

Tangela was asked to circle all of the expressions equivalent to $6(y+1)$.

Her work is shown below.

$6y+1$	$6y+7$	$6(y)+1(y)$	$6(y)+6(1)$	$6y+6$

1. Determine if Tangela correctly identified all of the equivalent expressions. If she is missing an expression, circle the expression(s) she is missing.

   No, Tangela identified two expressions that are not equivalent to $6(y+1)$.

2. If Tangela made an error, then explain her error and list the equivalent expressions.

   $6y + 1$ and $6(y) + 1(y)$ are not equivalent expressions to $6(y+1)$ as Tangela did not apply the distributive property correctly. If Tangela would have applied the distributive property correctly, she would have gotten then expression $6 \cdot y + 6 \cdot 1$, which is equivalent to:

   $6(y)+6(1)$ *and*
   $6y+6$

Consider allowing students who may struggle with written response to explain their answers verbally or record them on an app, like FlipGrid, for assessment.

Challenge early finishers to determine which properties were used to write each equivalent expression.